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[Ans to the question no. 1]

Add x_{21}, x_{22}, x_{23}

single cycle CPU:

$$T_{\text{single}} = 20 + 20 + 40 + 40 + 2 = 140 \text{ ns}$$

so, instruction time = 140 ns

pipelined CPU:

clock cycle time = maximum stage delay

$$T_{\text{pipe}} = \max(20, 20, 40, 40, 20) = 40 \text{ ns}$$

$$\begin{aligned} \text{instruction latency} &= 5 \text{ stages} \times 40 \text{ ns} \\ &= 200 \text{ ns} \end{aligned}$$

Thus, the single-cycle setup runs this instruction faster.

[Ans to the question no 2]

Ld X5, 10(X21)

- i. ALU SRC = 0
- ii. RegWrite = 1
- iii. MemRead = 1
- iv. MemWrite = 0
- v. MemtoReg = 0

[Ans to the question no 3]

Instruction and data memories are kept separate to avoid structural hazards in pipelined processors. This separation allows simultaneous instruction fetch and data access, improving performance and enabling parallel operations.

[Ans to the question no 4]

Yes, data hazards occur due to dependencies:

- i) $SD\ x22, 12(x21)$
- ii) $Add\ x21, x22, x23$
- iii) $LD\ x21, 12(x23)$
- iv) $Sub\ x21, x21, x24$
- v) $OR\ x22, x21, x26$
- vi) $LD\ x22, 0(x21)$

	cc1	cc2	cc3	cc4	cc5	cc6	cc7	cc8	cc9	cc10	cc11	cc12
i	F	D	Ex	Mem	WB							
ii		F	D	EX	Mem	WB						
iii			F	D	EX	Mem	WB					
				X	X	X	X	X				
iv					F	D	EX	Mem	WB			
v						F	D	EX	Mem	WB		
vi							X	X	X	Mem	WB	X
								F	D	EX	Mem	WB

$$\text{Avg. CPI} = \frac{\text{Total clock cycles}}{\text{Instruction count}}$$

$$= \frac{11 \ 12}{6}$$

$$\approx \cancel{183} = 2$$